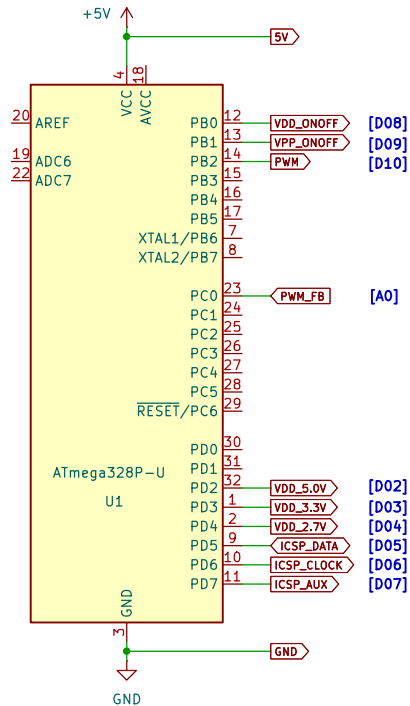


PicArd Programmer Hardware – Revision 2

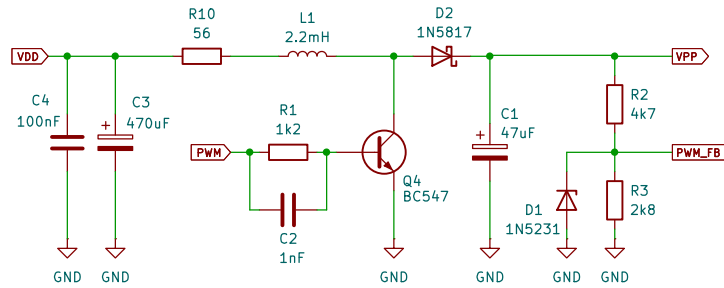
Arduino UNO R3 (ATMEGA328PU 32-lead TQFP)



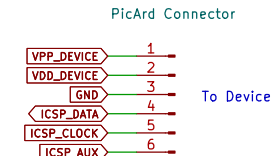
Notes:

- Showing only relevant connections and
- Into the brackets: Arduino Uno R3 board silk screen markings.

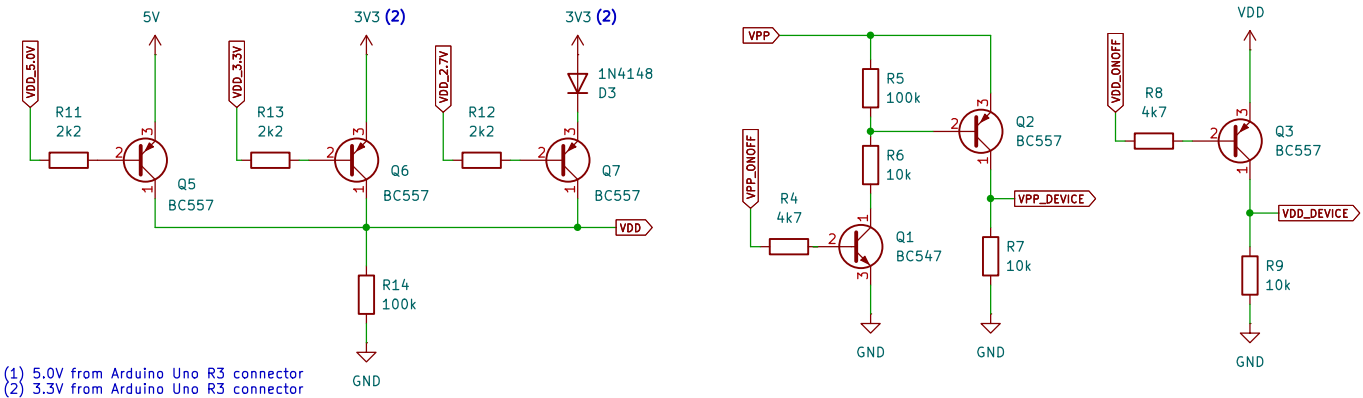
DC/DC Boost



Connector



VPP and VDD control



- 5.0V from Arduino Uno R3 connector
- 3.3V from Arduino Uno R3 connector

Table 1 – Labels Descriptions

Label	Description
5V	Arduino UNO R3 5V (from USB)
3V3	Positive 3.3V (from Arduino Uno)
PWM	Pulse with modulation to DC/DC Booster
PWM_FB	PWM voltage feedback
VDD_ONOFF	Turns on/off VDD to device
VPP_ONOFF	Turns on/off VPP to device
VDD_5.0V	Select/Unselect 5.0V as VDD
VDD_3.3V	Select/Unselect 3.3V as VDD
VDD_2.7V	Select/Unselect 2.7V as VDD
ICSP_DATA	In-circuit serial programming (ICSP) data pin
ICSP_CLOCK	ICSP clock pin
ICSP_AUX	ICSP for Low Voltage Programming (LVP)
VDD	Selected VDD
VDD_DEVICE	Device VDD Pin
VPP	Programming voltage
VPP_DEVICE	Device VPP Pin
GND	Ground

This is the current version of Programmer Hardware
Use this version in all new projects
Revision 2
<https://github.com/NelsonBittencourt/PicArd/>
Nelson Rossi Bittencourt

Sheet: /
File: picard_hardware.kicad_sch

Title: PicArd Programmer

Size: A4 Date: 2026-04-09

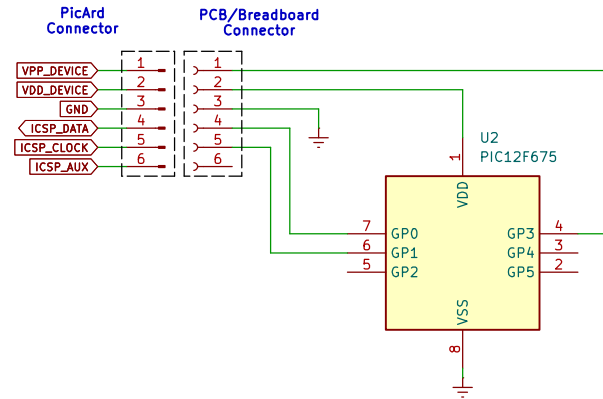
KiCad E.D.A. 10.0.0

Rev: 2

Id: 1/3

PicArd Programmer – Connection Examples

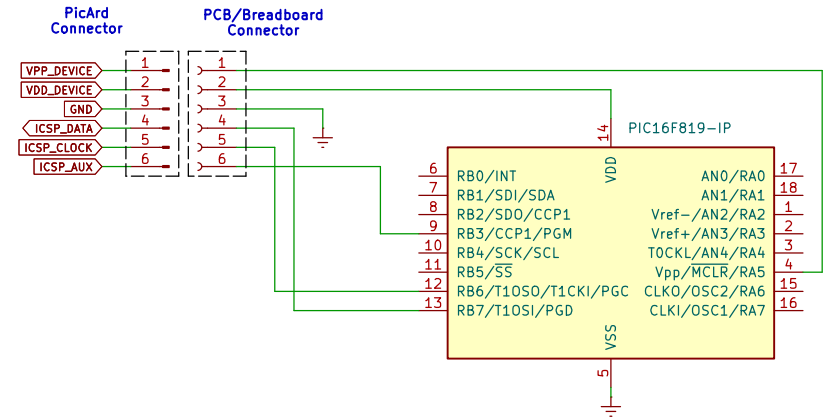
Example 1: PIC12F675 (PDIP, SOIC, DFN-S)



Notes:

- Connections according to "PIC12F629/675 Memory Programming Specification" (<https://ww1.microchip.com/downloads/en/DeviceDoc/41173c.pdf>);
- PIC12F675 can only be programmed via HVP, ICSP_AUX not used.

Example 2: PIC16F819 (DIP, SOIC)



Notes:

- Connections according to "PIC16F818/819 Memory Programming Specification" (<https://ww1.microchip.com/downloads/en/DeviceDoc/39603c.pdf>);
- PIC16F818/819 supports LVP and HVP modes. Selection is made via software.

Nelson Rossi Bittencourt

Sheet: /Picard_Connection_Example/

File: picard_hardware_connection_examples.kicad_sch

Title: PicArd Programmer – Connection Examples

Size: A4

Date: 2026-04-09

Rev: 2

KiCad E.D.A. 10.0.0

Id: 2/3

PicArd Programmer – Typical voltages and signals

Fig. A: PWM Signal

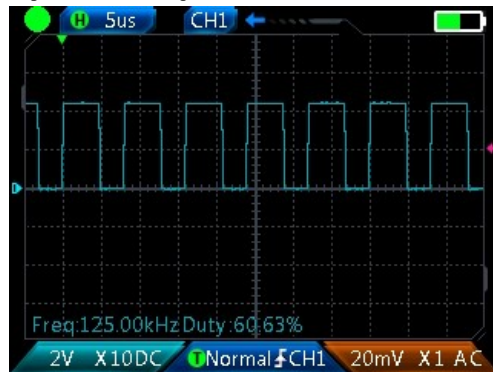


Fig. B: PWM in Q10 collector

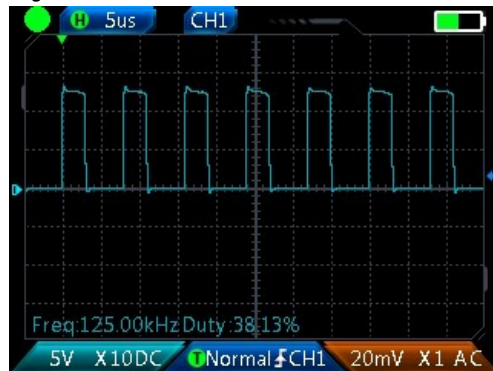
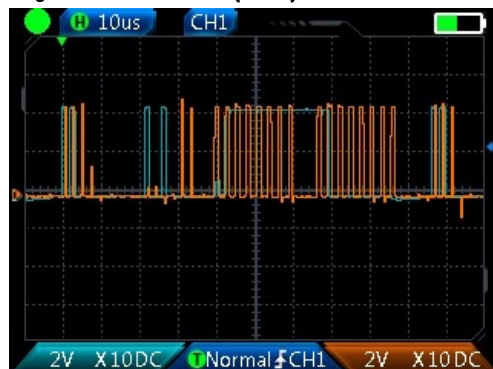
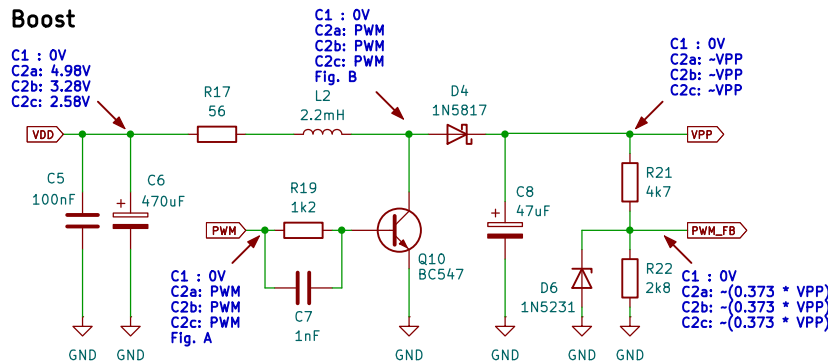


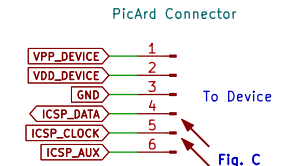
Fig. C: ICSP_DATA (blue) & ICSP_CLOCK



DC/DC Boost



Connector



VPP and VDD control

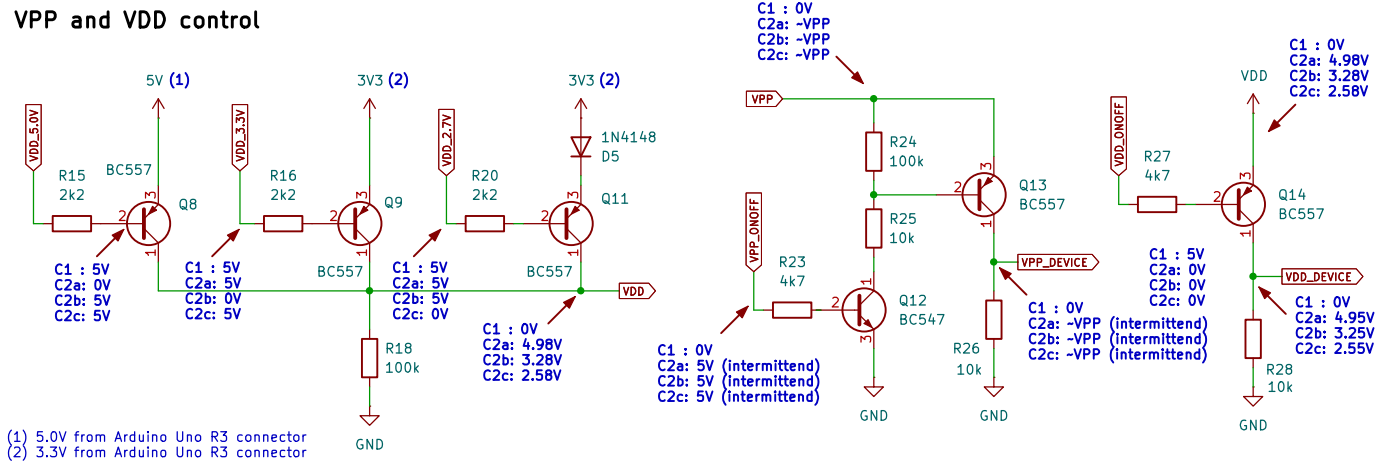


Table 2 – Condition Description	
Condition #	Description
C1	Arduino Uno connected in USB port/GUI connected or not
C2a	Data transfer for 5V PICs
C2b	Data transfer for 3.3V PICs
C2c	Data transfer for 2.7V PICs

This is the current version of Programmer Hardware
Use this version in all new projects
Revision 2
<https://github.com/NelsonBittencourt/PicArd/>
Nelson Rossi Bittencourt

Sheet: /
File: picard_voltages.kicad_sch

Title: PicArd Programmer – Typical voltages & signals

Size: A4
KiCad E.D.A. 10.0.0

Date: 2026-07-28

Rev: 0
Id: 3/3